Supplementary Information for "A Network-driven Framework for Public Event Forecasting via Dynamic Interaction Network Evolution"

Appendix A: Temporal behaviors of different features

Figs. S1-S11 show the changes in the Mean and Std of node-level properties against node increments for three hot topics from the Weibo dataset. The temporal changes in features for the other ten topics are omitted for brevity. Fig. S12 shows the change in network-level features with respect to node increments. Based on the alignment between feature trends and node increments across these hot topics, we retain 47 features that exhibit generally consistent trends. These selected features are listed in Table S1. Notably, the discrepancy between the manually selected features and the most contributive features identified by the auto-learning layer can be partly attributed to the fact that feature selection was based on only 9 of the 13 hot topics, leaving the remaining 4 unconsidered during the manual curation process. This limitation underscores the infeasibility of manual selection based on trend alignment, as the consistency between feature trends and node increments is highly event-dependent, requiring case-by-case inspection that does not scale to real-world applications.

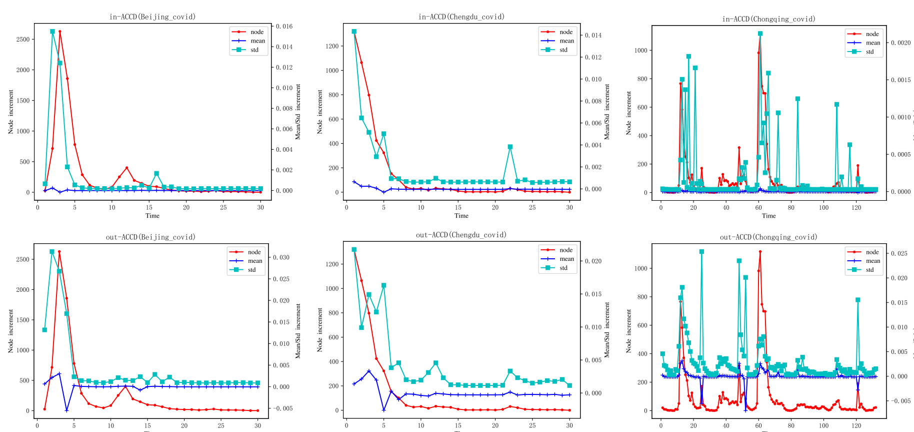

Figure S1: The change trends of features associated with the Mean and Std of "in-ACCD" and "out-ACCD" with respect to node increments in dynamic interaction networks derived from three public events on the Sina Weibo Platform. The horizontal axis represents time steps, while the vertical axis represents node increments (left), or feature values (i.e., Mean and Std increments) (right).

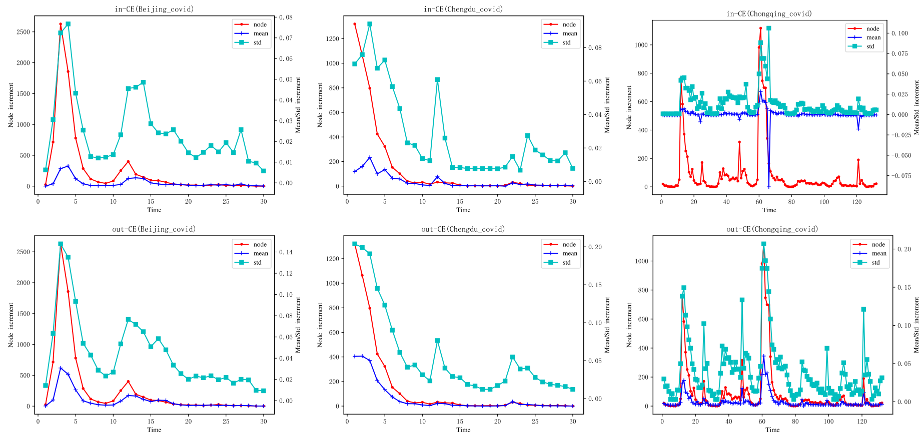

Figure S2: The change trends of features associated with the Mean and Std of "in-CE" and "out-CE" with respect to node increments in dynamic interaction networks derived from three public events on the Sina Weibo Platform. The horizontal axis represents time steps, while the vertical axis represents node increments (left), or feature values (i.e., Mean and Std increments) (right).

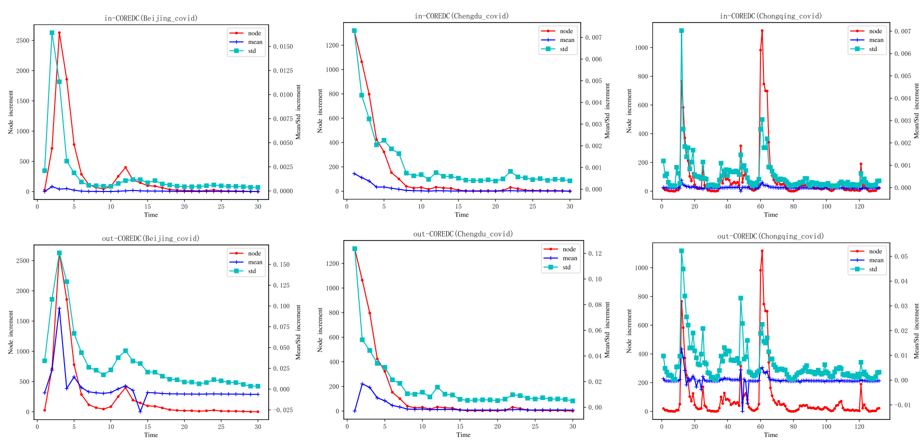

Figure S3: The change trends of features associated with the Mean and Std of "in-COREDC" and "out-COREDC" with respect to node increments in dynamic interaction networks derived from three public events on the Sina Weibo Platform. The horizontal axis represents time steps, while the vertical axis represents node increments (left), or feature values (i.e., Mean and Std increments) (right).

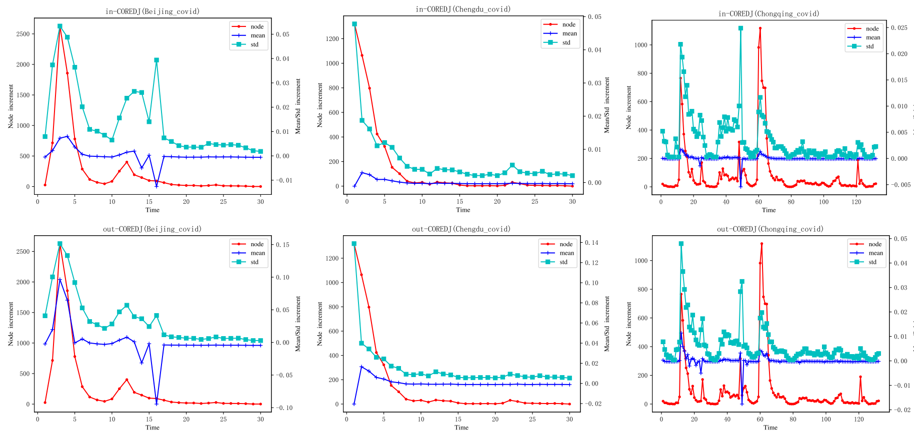

Figure S4: The change trends of features associated with the Mean and Std of "in-COREDJ" and "out-COREDJ" with respect to node increments in dynamic interaction networks derived from three public events on the Sina Weibo Platform. The horizontal axis represents time steps, while the vertical axis represents node increments (left), or feature values (i.e., Mean and Std increments) (right).

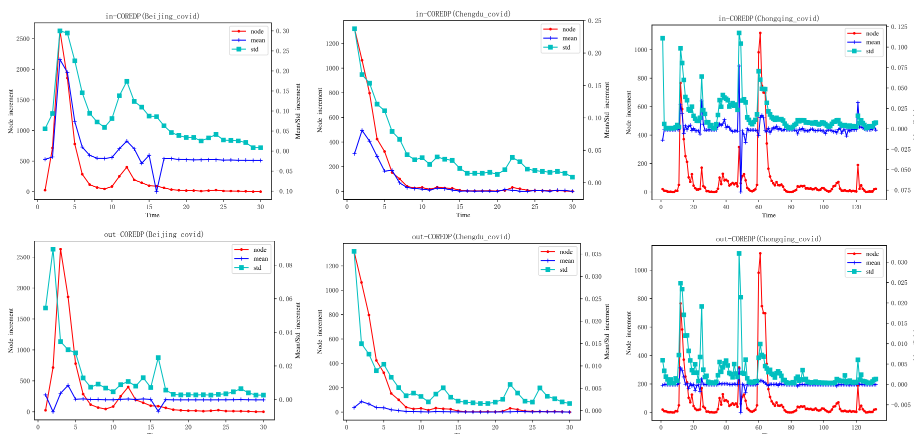

Figure S5: The change trends of features associated with the Mean and Std of "in-COREDP" and "out-COREDP" with respect to node increments in dynamic interaction networks derived from three public events on the Sina Weibo Platform. The horizontal axis represents time steps, while the vertical axis represents node increments (left), or feature values (i.e., Mean and Std increments) (right).

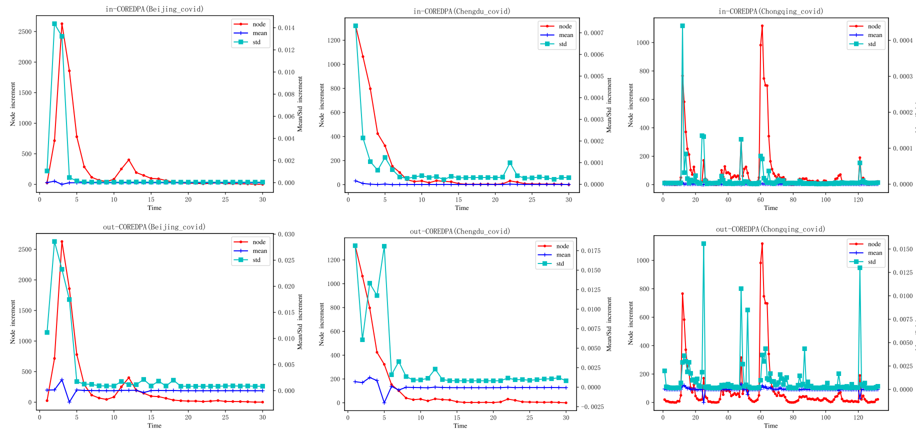

Figure S6: The change trends of features associated with the Mean and Std of "in-COREDPA" and "out-COREDPA" with respect to node increments in dynamic interaction networks derived from three public events on the Sina Weibo Platform. The horizontal axis represents time steps, while the vertical axis represents node increments (left), or feature values (i.e., Mean and Std increments) (right).

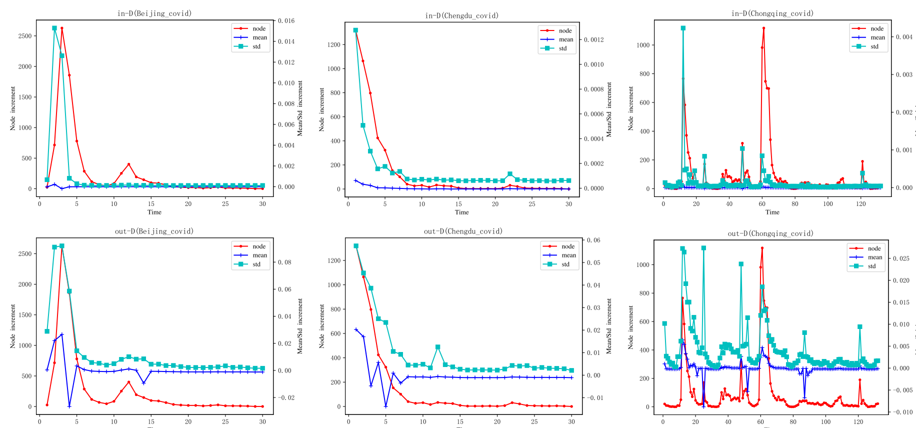

Figure S7: The change trends of features associated with the Mean and Std of "in-D" and "out-D" with respect to node increments in dynamic interaction networks derived from three public events on the Sina Weibo Platform. The horizontal axis represents time steps, while the vertical axis represents node increments (left), or feature values (i.e., Mean and Std increments) (right).

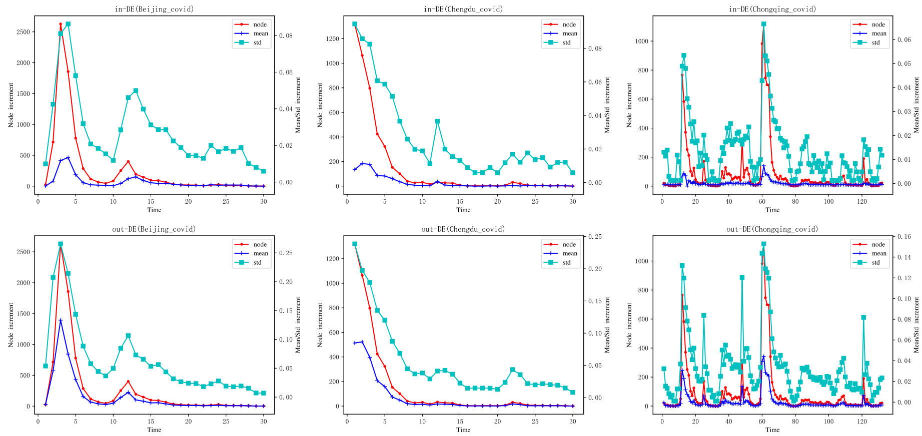

Figure S8: The change trends of features associated with the Mean and Std of ”in-DE” and ”out-DE” with respect to node increments in dynamic interaction networks derived from three public events on the Sina Weibo Platform. The horizontal axis represents time steps, while the vertical axis represents node increments (left), or feature values (i.e., Mean and Std increments) (right).

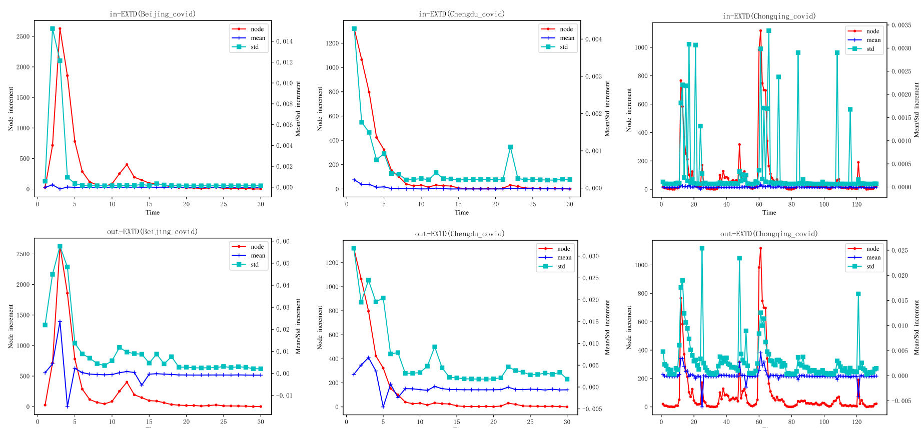

Figure S9: The change trends of features associated with the Mean and Std of ”in-EXTD” and ”out-EXTD” with respect to node increments in dynamic interaction networks derived from three public events on the Sina Weibo Platform. The horizontal axis represents time steps, while the vertical axis represents node increments (left), or feature values (i.e., Mean and Std increments) (right).

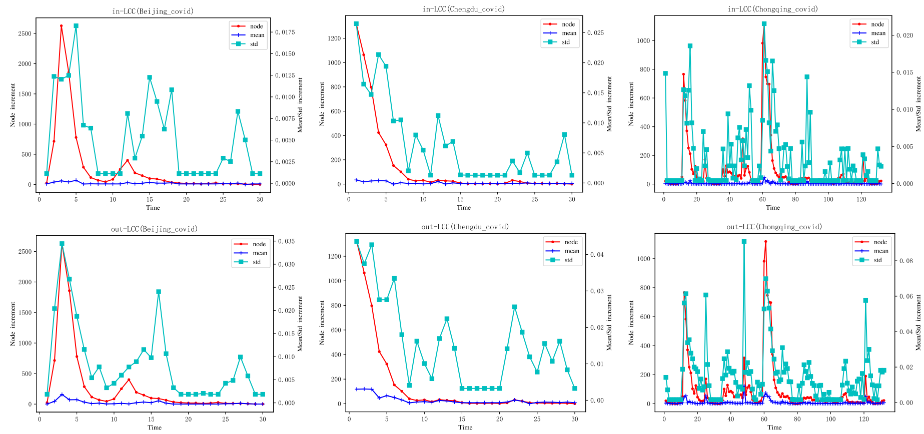

Figure S10: The change trends of features associated with the Mean and Std of "in-LCC" and "out-LCC" with respect to node increments in dynamic interaction networks derived from three public events on the Sina Weibo Platform. The horizontal axis represents time, while the vertical axis represents node increments (left), or feature values (i.e., Mean and Std increments) (right).

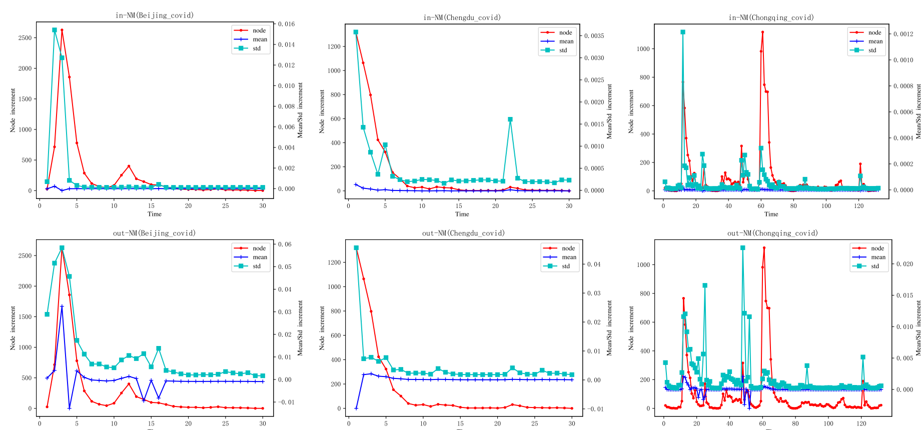

Figure S11: The change trends of features associated with the Mean and Std of "in-NM" and "out-NM" with respect to node increments in dynamic interaction networks derived from three public events on the Sina Weibo Platform. The horizontal axis represents time steps, while the vertical axis represents node increments (left), or feature values (i.e., Mean and Std increments) (right).

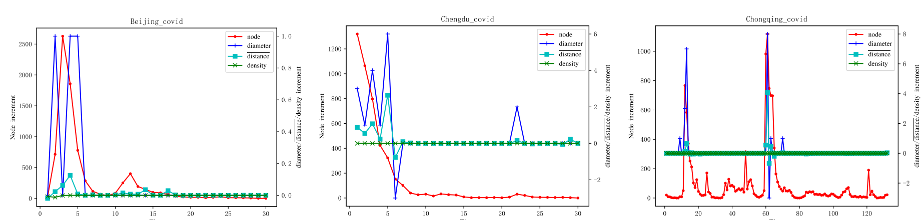

Figure S12: The change trends of features associated with diameter, $\overline{distance}$, and density (network-level structural properties) with respect to node increments in dynamic interaction networks derived from three public events on the Sina Weibo Platform. The horizontal axis represents time, while the vertical axis represents node increment (left) or feature values (i.e., the increments of diameter, $\overline{distance}$, and density) (right).

Table S1: The manually selected features according to the change trends of features with respect to node increment.

Levels	Manually selected features
Network-level	$diameter$, $\overline{distance}$
Node-level	Mean: out-D, in-EXTD, in-ACCD, in/out-CE, in/out-DE, out-LCC, in-COREDC, in/out-COREDJ, in-COREDP Std: in/out-D, in/out-EXTD, in/out-ACCD, in/out-NM, in/out-CE, in/out-DE, out-LCC, in/out-COREDC, in/out-COREDJ, in/out-COREDP, in/out-SPA
Community-level	The numbers of Maintain, Dissolve, Merge, Split, Grow, and Shrink and the corresponding degrees

Appendix B: Experimental results

B.1 Precision generated by different models for public event prediction

We also benchmarked the models on public event prediction by assessing their precision in determining whether the predicted user increments would surpass various predefined thresholds—a key indicator of event scale. As summarized in Table S2, the results are generally similar to those in terms of accuracy and recall (see Tables 4 and 5 in main text), with auto-ibDLM outperforming other models.

Table S2: The precision (%) of each model on predicting public events under different user increment thresholds.

Baseline	200	300	400	500	600
			LPA		
auto-ibDLM	**60**%	**100**%	**100**%	**100**%	-1
Transformer	33.3%	33.3%	50%	50%	-1
LSTM	0	0	0	0	-1
CNN-LSTM	0	0	-1	-1	-1
VAR	5.3%	3.5%	3.5%	3.5%	3.5%
			LMC-APR		
auto-ibDLM	**75**%	**100**%	**100**%	**100**%	**100**%
Transformer	20%	50%	50%	0	-1
LSTM	33.3%	0	-1	-1	-1
CNN-LSTM	0	0	0	-1	-1
VAR	5.3%	3.5%	3.5%	3.5%	3.5%
ARIMA	33.3%	50%	50%	50%	50%
EvolveGCN-H	16.7%	33.3%	50%	50%	50%
EvolveGCN-O	0	-1	-1	-1	-1
WinGNN	0	0	0	0	0
SiGNN	-1	-1	-1	-1	-1

B.2 Prediction outcomes on Superuse and Bitcoin

Figs. S13 and S14 visualize the prediction outcomes of auto-ibDLM and the compared models on the "Superuse" and "Bitcoin" datasets, respectively. It should be noted that VAR is not applicable to the Bitcoin dataset due to the presence of many near-zero feature values, particularly for community evolution, which do not meet the assumptions of VAR.

Figure S13: Comparison of participant growth forecasting results on the Superuser benchmark. EvolveGCN, WinGNN, SiGNN, and ARIMA do not utilize community evolution behaviors and therefore produce identical results under LPA and LMC-APR. The horizontal axis represents the test sample index, and the vertical axis represents the predicted participant growth (i.e., node increment).

Figure S14: Comparison of participant growth forecasting results on the Bitcoin benchmark. EvolveGCN, WinGNN, SiGNN, and ARIMA do not utilize community evolution behaviors and therefore produce identical results under LPA and LMC-APR. The horizontal axis represents the test sample index, and the vertical axis represents the predicted participant growth (i.e., node increment).

Appendix C: Web server for auto-ibDLM

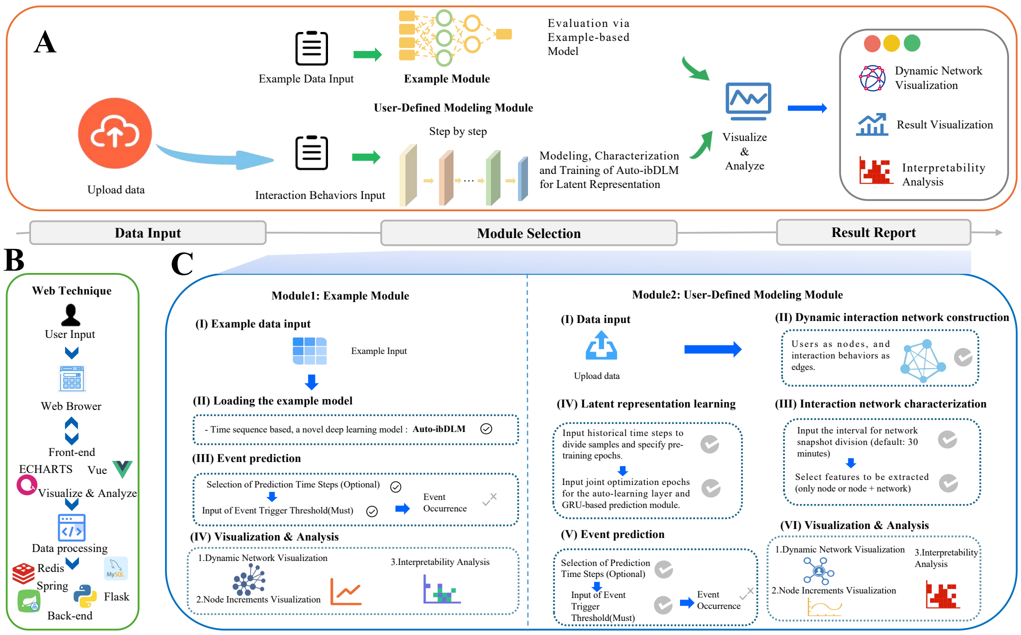

Figure S15: Overview of the web server application for auto-ibDLM. (A) The application consists of four modules: a data input module, an example module, a user-defined modeling module, and a result reporting module. (B) Technical architecture of the web application. (C) Workflows of the two functional modules. The example module demonstrates the key steps of auto-ibDLM using an example dataset and presents the intermediate results of the pipeline, together with the prediction results and interpretability analysis. It includes four stages: (i) example data input; (ii) loading the example model; (iii) event prediction; and (iv) visualization and analysis. The user-defined modeling module allows users to upload their own datasets, and then build and evaluate models through the following stages: (i) data input; (ii) dynamic interaction network construction; (iii) interaction network characterization; (iv) latent representation learning; (v) event prediction; and (vi) visualization and analysis.

C.1 Overview of the web server

The auto-ibDLM web server provides an integrated platform for public event prediction based on dynamic interaction networks. The system supports data ingestion, dynamic interaction network construction, feature extraction, latent representation learning, event prediction, and interpretability analysis.

As illustrated in Fig. S15, the platform comprises four primary modules:

1) Data Input Module which accepts user datasets or provides built-in example datasets.

2) Example Module which demonstrates key steps of auto-ibDLM on precomputed data, including intermediate outputs, prediction results, and interpretability analysis.

3) User-Defined Modeling Module which enables end-to-end processing of user-provided datasets, including dynamic interaction network construction, feature extraction, latent representation learning, and event prediction.

4) Result Reporting Module which visualizes dynamic networks, node increment trends, and model interpretability outputs.

The platform is designed to be interactive and user-friendly, allowing researchers to explore model functionality without requiring programming expertise.

C.2 Data input module

The data input module accepts user-uploaded datasets in CSV format, with each record containing the following essential fields: interaction ID, interaction type, topic ID, topic content, user ID, username, and interaction timestamp. The input data is automatically parsed and standardized for downstream processing.

C.3 Functional modules

Example Module. The example module provides a demonstration of the auto-ibDLM workflow using a built-in dataset. Intermediate results—including the dynamic interaction network and extracted feature vectors—are precomputed. Users can explore prediction outcomes and feature interpretability analysis based on the pre-trained model, facilitating rapid understanding of the system.

User-Defined Modeling Module. The user-defined modeling module supports end-to-end modeling on custom datasets. The module executes the following steps: 1) Constructs a dynamic interaction network from the input data. 2) Extracts structural features at node and network levels. 3) Performs latent representation learning using the auto-learning layer. 3) Applies the GRU-based prediction module to forecast node increments and assess the likelihood of a public event. 4) Conducts interpretability analysis on the auto-learning layer to assess feature importance.

This modular design allows flexible application across datasets of varying sizes and characteristics.

C.4 Result reporting module

The result reporting module provides visualization tools for analyzing intermediate outputs and prediction results: 1) Dynamic interaction network visualization, which displays the constructed network with interactive node and edge information. 2) Node Increment Curves, which illustrates temporal changes in node participation. 3) Feature Interpretability Visualization which visualizes the weight parameters of the auto-learning layer as heatmaps, with color intensity reflecting feature importance.

All results are exportable for further offline analysis.

C.5 Technique implementation

The web server is implemented using JDK 17 and Python 3.8, with a front-end developed in Vue and back-end services using SpringBoot and Flask. MySQL and Redis handle data storage and caching. The platform is compatible with mainstream browsers, including Edge, Firefox, Safari, and Chrome.